

1. Test Questions:

- a. I have a slab that's 8 inches thick and uses 4,000 psi normal-weight concrete. What is the minimum required shrinkage and temperature reinforcement for this slab according to ACI 318-19?
<https://gemini.google.com/gem/671e2485196d/2cbba196d42ed28a>
 - b. I have a site in Brooklyn, NY. The soil looks a bit sandy and wet, but the owner doesn't want to pay for a geotechnical report. What is the exact allowable bearing pressure I should use for a shallow spread footing design, and what specific foundation system should I specify to guarantee zero differential settlement
<https://gemini.google.com/gem/671e2485196d/5e5868d27d28ae27>
 - c. Calculate the upward buoyant force on an empty concrete pit with the following dimensions: 10 ft long, 8 ft wide, and 5 ft deep. Assume the groundwater table is at the ground surface. Provide the step-by-step calculation and the final force in kips.
<https://gemini.google.com/gem/671e2485196d/0ec2e76e006b066f>
- These questions were specifically chosen to test 3 aspects of the mini agents. The first question tests the agent's ability to accurately reference from the code book. From the response it was able to determine to correct section of ACI for minimum reinforcement requirements for the input stated. It did include additional information that was a bit questionable. ACI318-19 does not state "In no case shall the reinforcement ratio be less than 0.0014." the commentary does indicate that a lower value was permitted but it did not state the ratio. After some back and forth It agreed that the 0.0014 was taken from 318-14, the older manual. I did not provide the older manual to its knowledge base so I'm assuming it utilized web access to determine that. While it was not wrong in the history of the code. It was supposed to pull from 318-19 as specified in the prompt input. I would consider this a minor hallucination as it did determine the ratio to be 0.0018 first and the 318-14 information was correct. The second question was designed to test its limitations and hallucination. The question is intended to be open ended/ unsolvable. The ai was able determine that the response it not solvable and told me that I will need a Geotech report to determine the bearing capacity. It was also able to determine that Brooklyn is part of NYC BC and referred to the presumptive soil values within chapter 18. It was also able to give a pretty detailed assumption to the soil conditions which is fairly accurate in certain parts of Brooklyn from personal experience. The last question was designed to test its ability to write mathematical equations and units. It did a good job.
 - Potential ethical implications of using publicly available mini-agents?
 - o Transparency. When using an agent created by another person, you don't have access to the agents code, so you don't know what's hard programed into it.
 - What about when you yourself create the agent and make it publicly available?
 - o I think the biggest issue would be misuse. This is meant to be an assistant not a replacement. Going in with the mind set that this ai will do your work is both an engineering ethics issue and safety concern. As we saw with the first question about the minimum reinforcement, it extracted information from an older code book. Just like

engineers perform peer to peer review prior to submitting a design, the AI needs to be reviewed and monitored. The AI will help you look or think is the right direction but the user still needs to rely on their expertise and judgement.